

Student Enrollment & Course Management System

(SECMS)

A Salesforce Application Project

Team Members

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1. Project Overview

1.1 About SECMS

SECMS is a Salesforce application for an educational institution. It manages students, courses, enrollments, instructors, approvals, fees and reports. All important academic information is kept in one system, helping staff work faster and reduce manual work.

1.2 Purpose

Replace separate manual enrollment work with one Salesforce system.

Main goals

- Keep student data correct
- Organize enrollment work
- Automate approvals and checks
- Create clear reports

1.3 Problem Statement

Enrollment officers see scattered data and do many tasks manually. Administrators may receive reports late because information is not in one place.

2. Ideation and Analysis

2.1 User Persona: Enrollment Officer

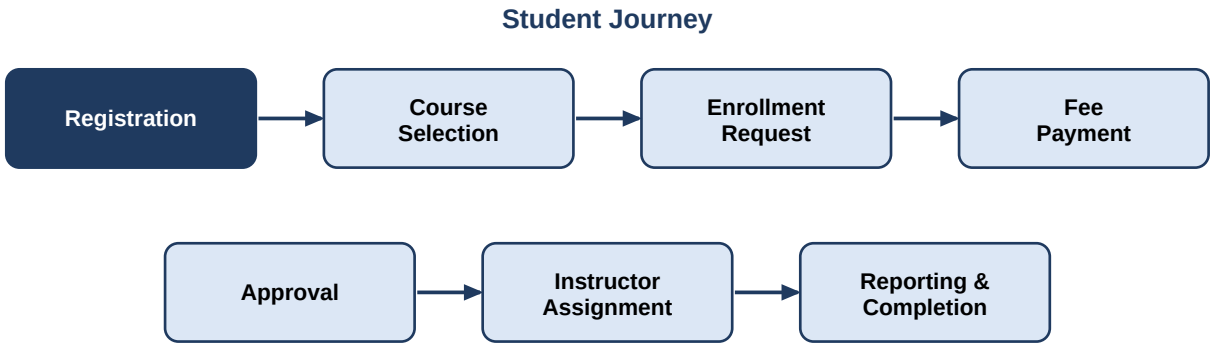
- **Think and feel:** worried about duplicate bookings and loss of money
- **See:** many spreadsheets and emails
- **Do:** check data again and follow up manually
- **Problems:** duplicate records and slow approvals
- **Needs:** automation, dashboards and faster processing

2.2 Brainstormed Ideas

Ideas considered: spreadsheet automation, a separate enrollment portal, ERP connection, and a Salesforce system. These were grouped as manual, partly automatic, and fully automatic solutions. Salesforce SECMS was selected during project planning as the main solution.

2.3 Student Journey

Registration -> Course Selection -> Enrollment Request -> Fee Payment -> Approval -> Instructor Assignment -> Reporting and Completion. These steps describe the complete enrollment process.



3. Requirement Analysis

3.1 Functional Requirements

ID	Requirement
FR-1	Manage student records
FR-2	Create and manage courses
FR-3	Request, approve or reject enrollments
FR-4	Automatically assign instructors
FR-5	Create reports and dashboards

3.2 Non-Functional Requirements

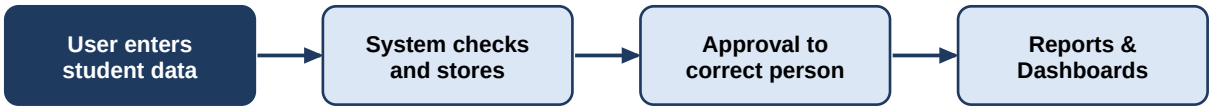
- **Usability:** screens and forms should be simple and easy to use (e.g., quick registration, one-click approval, role-based dashboards)
- **Security:** only authorized users should see or change sensitive data (e.g., protected student records)
- **Reliability:** enrollment and approval processes should work correctly
- **Performance:** the system should respond quickly, especially during admission time (e.g., quick search, quick dashboards)
- **Scalability:** it should support more students, courses, departments and users later
- **Availability:** the system should be available during working hours with reduced downtime (e.g., regular backup)

4. Solution and Technical Architecture

4.1 Data Flow

- The user enters student and enrollment data
- The system checks the data and stores it
- Approval sends the enrollment to the correct person
- Reports and dashboards use the stored data

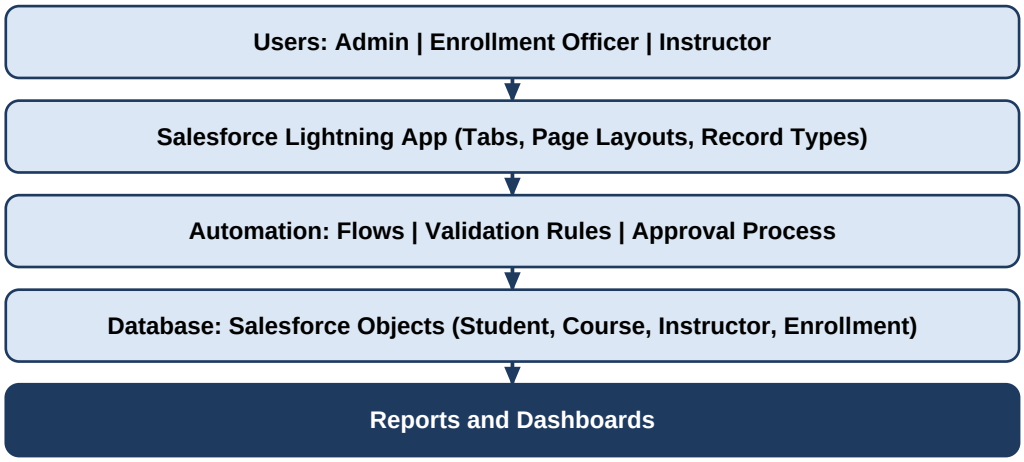
Data Flow



4.2 Technology Stack

Layer	Technology
Platform	Salesforce Lightning
Database	Salesforce Objects
Logic layer	Salesforce Flows
Automation	Validation Rules and Approval Process
Reports	Salesforce Reports and Dashboards

SECMS Architecture



4.3 Solution Summary

- **Problem:** manual enrollment takes time
- **Solution:** Salesforce-based SECMS
- **Main idea:** automate the process from enrollment to reporting
- **Benefit:** better student service and support for multiple departments

5. Project Planning and Scheduling

5.1 Agile Methodology

- Agile working method with sprint-based execution
- Work is divided into Epics, User Stories and Story Points
- Velocity is used to estimate the amount of work completed
- Sprint-1 starts with Salesforce setup and data modeling

5.2 Sprint Planning

User stories cover Salesforce setup, objects, tabs, fields, relationships, page layouts and automation. Important objects are Student, Course, Enrollment and Instructor. Each task has story points, priority and a team member; high-priority tasks are completed first. Several sprints cover setup, data modeling, automation, security, reports and dashboards.

5.3 Project Milestones

- Create Salesforce account
- Create objects, tabs and app
- Create fields and relationships
- Customize layouts
- Add validation rules
- Create flows and approval process
- Create reports, dashboards and security

5.4 Learning Outcomes

Build objects, relationships, validation rules, automation, approvals, reports, dashboards and security.

6. Salesforce Environment Setup

6.1 What is Salesforce?

Salesforce is a cloud platform used to manage data, work and customer relationships. In this project it is used to build the student enrollment application. Users can store data securely in the cloud and work from the Salesforce interface.

6.2 Developer Account Creation

- Enter name, email, role, college/company, country (India), postal code and username
- Username must follow the format username@organization.com (need not be a real email)
- Click Sign Me Up and activate the account from the email
- Verify email, create a password and answer the security question
- Salesforce then opens the Setup page, completing activation

7. Implementation

Milestone 2: Objects

Student Object

Purpose: store student personal details, contact information and enrollment status. Steps: Setup -> Object Manager -> Create -> Custom Object; set Label and Object Name as Student; use Student Name as the record name; enable reports and search as needed, then save. Course, Instructor and Enrollment objects are created the same way.

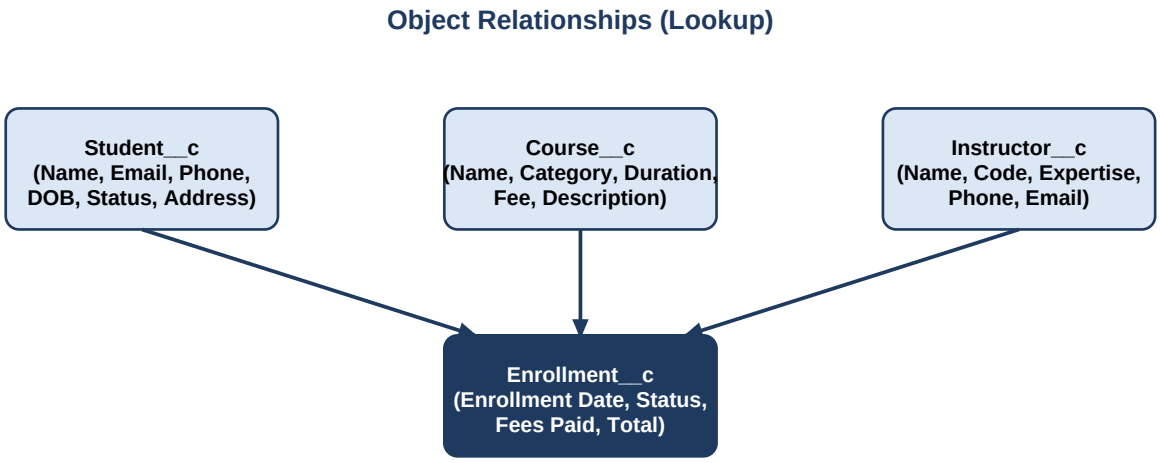
Milestone 3: Fields

Object	Fields
Student	Student Name, Email, Phone, Date of Birth, Student Status, Address
Course	Course Name, Category, Duration, Course Fee, Description
Instructor	Instructor Name, Instructor Code, Expertise (Picklist), Phone, Email
Enrollment	Student, Course, Instructor, Enrollment Date, Enrollment Status, Fees Paid (Checkbox), Total (Currency), Comment (Text Area)

Milestone 4: Relationships

Lookup Relationships are created from Enrollment__c via Setup -> Object Manager -> Enrollment__c -> Fields & Relationships -> New.

- **Enrollment to Student:** Lookup to Student__c, label Student
- **Enrollment to Course:** Lookup to Course__c, label Course (links every enrollment with its course)
- **Enrollment to Instructor:** Lookup to Instructor__c, label Instructor (stores the assigned instructor)



For each field, click Next through Field-Level Security and Layout options, ensure it is visible and added to the layout, then Save.

Milestone 5: Validation Rules

Rule 1: Student Email Domain Validation

Allows only the required college/student email domain. Object: Student__c. Rule name: Student_Email_Domain_Validation. Example formula: NOT(CONTAINS(Email__c, "@student.edu")). An error is shown when the email is not valid.

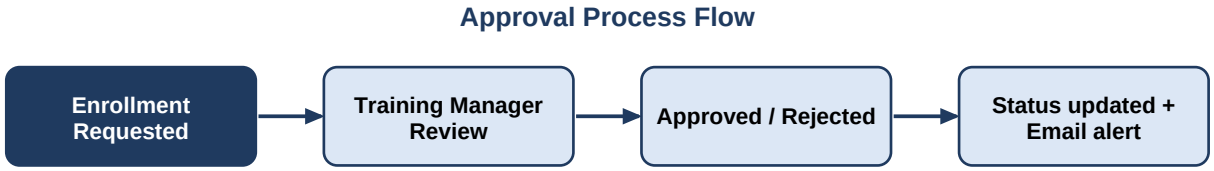
Rule 2: Fees Must Be Paid Before Approval

Object: Enrollment__c. Rule name: Fees_Must_Be_Paid_Before_Approval. Error condition: Enrollment_Status__c = Approved and Fees_Paid__c = FALSE. Error message: fees must be paid before approval.

Milestone 6: Approval Process

- Create the approval process for Enrollment__c using Standard Setup

- Process name: Approval_Enrollment_Request
- Entry condition: Enrollment_Status__c = Requested
- Add an approval step with the Training Manager as approver
- Final approval action: field update sets Enrollment_Status__c to Approved
- Final rejection action: field update sets Enrollment_Status__c to Rejected, plus an email alert to the student
- Activate the approval process after checking all settings



Milestone 7: Tabs

Setup -> Tabs -> Custom Object Tabs -> New. Select Student, choose a tab style, assign to all profiles and save. Tabs for Course, Instructor and Enrollment are created the same way.

Milestone 8: Lightning App

Setup -> App Manager -> New Lightning App. App name: Student Enrollment & Course Management; standard navigation style. Give access to Admin, Enrollment Officer and Instructor profiles. Add Student, Course, Instructor, Enrollment, Reports, Dashboards, Home and Tasks, then save.

Milestone 9: Page Layouts and Record Types

Record Types for Enrollment

Purpose: separate New Enrollment and Re-Enrollment processes. Go to Enrollment__c -> Record Types -> New and create two record types, enabled for the required profiles (Admin, Enrollment Officer) with the correct page layout.

Page Layouts

- **New Enrollment layout:** Student, Course, Instructor, Enrollment Date, Status, Fees; unneeded fields removed
- **Re-Enrollment layout:** previous enrollment reference, re-enrollment reason, status, fees
- Assign each layout to the correct record type and profile

Enrollment Related Lists on Parent Objects

Add the Enrollments related list to the Student, Course and Instructor page layouts so users can see enrollments from each parent record. Save each layout.

Milestone 10: Automation with Flow

Activity 1: Auto-fill Enrollment Date

Record-Triggered Flow on Enrollment__c, triggered when a record is created. Update Records sets Enrollment_Date__c to $Flow.CurrentDate. Saved as SetEnrollmentDate and activated.

Activity 2: Send Email When Enrollment Is Approved

Create an email template and email alert. Record-Triggered Flow on Enrollment__c (on update) with condition Enrollment_Status__c = Approved: get the related Student record, send the email alert to the student, notify the administrator if needed, and set Student__c.Status__c to Active. Saved as Enrollment Approved Action and activated.

Activity 3: Auto-Assign Instructor by Course Category

Course Category	Instructor
Technical	Instructor A (INSTR_A)
Language	Instructor B (INSTR_B)
Non-Technical	Instructor C (INSTR_C)

Record-Triggered Flow on Enrollment__c (created/updated): Get Records for the related Course, a Decision element on Course__r.Category__c, then get the correct instructor and update Instructor__c.

Activity 4: Auto-Create Follow-Up Task

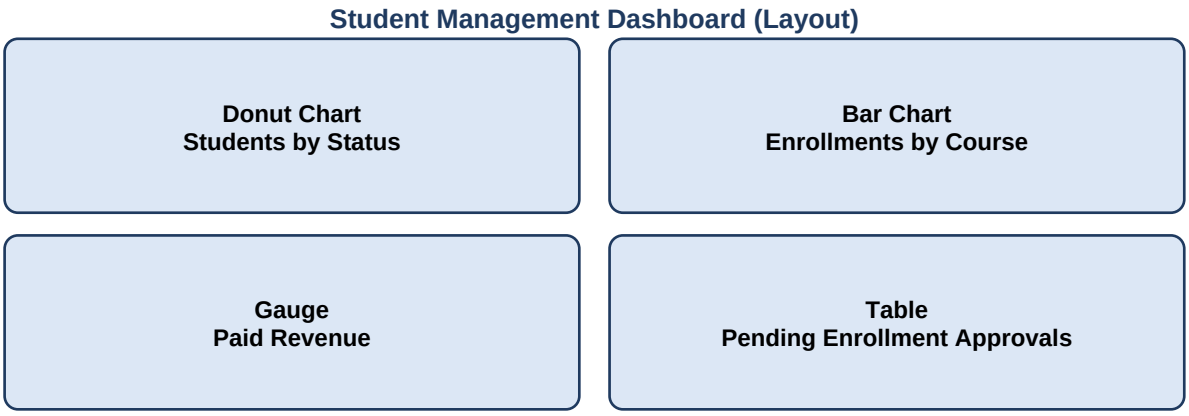
Record-Triggered Flow on Enrollment__c when Enrollment_Status__c = Requested. Get the admin user, then Create Records for a Task: Subject "Follow up on enrollment request", OwnerId = Admin user, WhatId = Enrollment Id, ActivityDate = TODAY() + 2. Saved as Create_Followup_Task and activated.

Milestone 11: Reports

Report	Details
Students by Status	Report type Students; group rows by Status__c; show key contact columns
Enrollments by Course	Report type Enrollments; group by Course__r.Name; show record count per course
Pending Enrollment Approvals	Filter Enrollment_Status__c = Requested; columns Student, Course, Owner, Created Date
Revenue Report (Fees Paid)	Filter Fees_Paid__c = TRUE; group by course to show fee totals

Milestone 12: Dashboard

Dashboards -> New Dashboard, named Student Management Dashboard, saved in a new dashboard folder. Components:



- Donut chart: Students by Status
- Bar chart: Enrollments by Course
- Gauge: paid revenue (sum of fees paid, from Revenue report)
- Table: Pending Enrollment Approvals

Milestone 13: Security Setup

Profiles

- **Admin:** full access
- **Enrollment Officer:** Enrollment__c Read, Create, Edit; Student__c / Course__c Read (Edit if required)
- **Instructor:** read-only access to assigned enrollments
- Permission Sets can give extra access when needed

Sharing Settings and Organization-Wide Defaults

- Students (Student__c): Public Read/Write
- Courses (Course__c): Public Read/Write
- Enrollments (Enrollment__c): Private
- Instructor access to assigned enrollments (optional): add a User lookup on Enrollment (Instructor User)

8. Testing

Testing is done at the end. It checks student, course, instructor and enrollment creation, and verifies validation rules, record types, flows, approvals and reports.

9. Advantages and Disadvantages

9.1 Advantages

- Data is kept in one place

- Workflows are automatic
- Reports are available quickly

9.2 Disadvantages

- The project depends on Salesforce
- Needs setup work

10. Conclusion and Future Scope

10.1 Conclusion

SECMS makes the student academic process easier, from registration to course enrollment, instructor assignment, approval, fee tracking and reporting. Salesforce validation rules, flows, approvals and dashboards help reduce manual work and improve data accuracy.

10.2 Future Scope

- Payment gateway
- Student self-service portal
- Better analytics
- Mobile access

11. Appendix

- Salesforce configuration / source code
- Salesforce object dataset
- Developer-org / demo